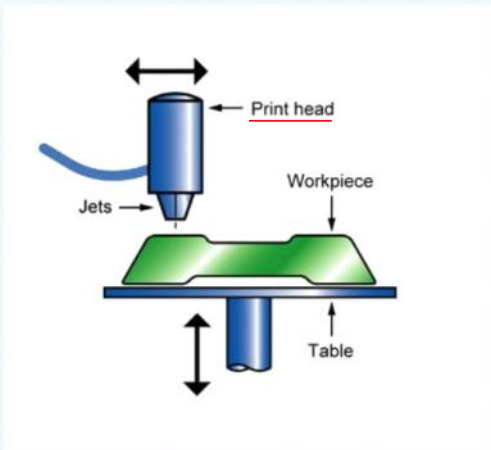


Week 6 - Quiz

21 May 2025 14:25

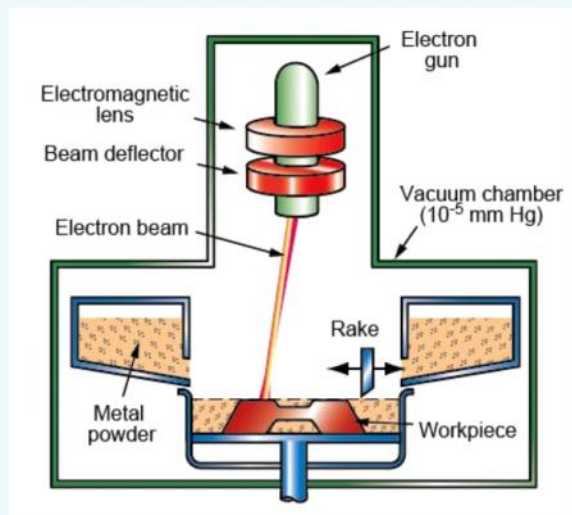
Which process is this?



Select one:

- ☐ a. Laminated object manufacturing
- ☒ b. 3D printing
- ☐ c. Selective laser sintering
- ☐ d. Laser powder forming
- ☐ e. Selective laser melting
- ☐ f. Electron beam melting
- ☐ g. Stereolithography

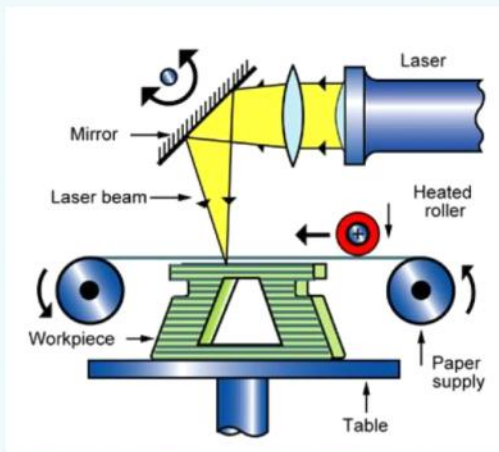
Which process is this?



Select one:

- ☐ a. Laser powder forming
- ☐ b. Selective laser sintering
- ☐ c. Selective laser melting
- ☐ d. Laminated object manufacturing
- ☐ e. 3D printing
- ☒ f. Electron beam melting
- ☐ g. Stereolithography

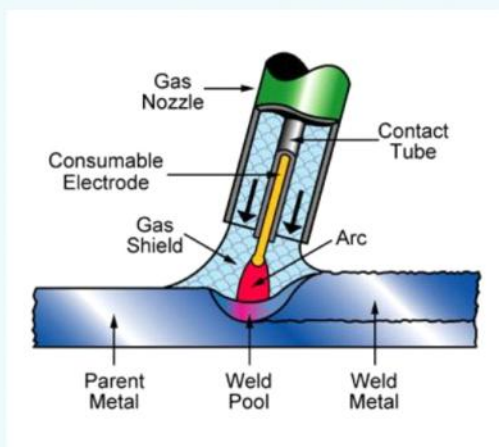
Which process is this?



Select one:

- ☐ a. Selective laser melting
- ☐ b. 3D printing
- ☐ c. Stereolithography
- ☐ d. Laser powder forming
- ☐ e. Electron beam melting
- ☐ f. Selective laser sintering
- ☒ g. Laminated object manufacturing

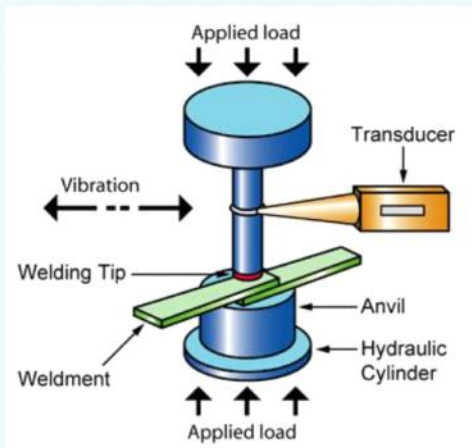
Which process is this?



Select one:

- ☐ a. Oxyacetylene welding
- ☒ b. Gas metal arc welding
- ☐ c. Ultrasonic welding
- ☐ d. Gas tungsten arc welding
- ☐ e. Power beam welding
- ☐ f. Friction Stir Welding
- ☐ g. Brazing

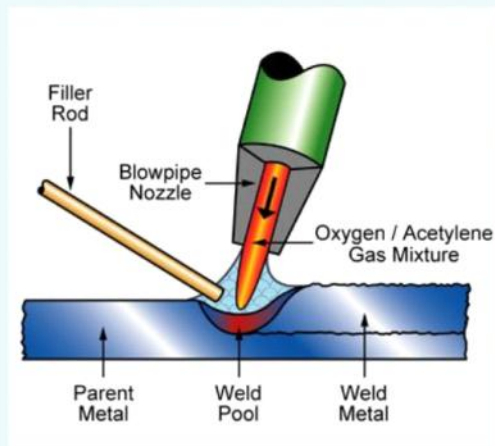
Which process is this?



Select one:

- ☐ a. Oxyacetylene welding
- ☐ b. Friction Stir Welding
- ☐ c. Gas metal arc welding
- ☐ d. Power beam welding
- ☐ e. Brazing
- ☐ f. Gas tungsten arc welding
- ☒ g. Ultrasonic welding

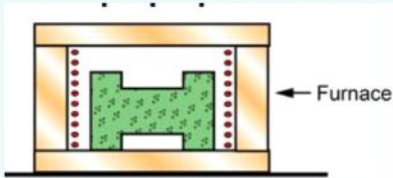
Which process is this?



Select one:

- ☐ a. Ultrasonic welding
- ☐ b. Brazing
- ☐ c. Gas metal arc welding
- ☒ d. Oxyacetylene welding
- ☐ e. Gas tungsten arc welding
- ☐ f. Friction Stir Welding
- ☐ g. Power beam welding

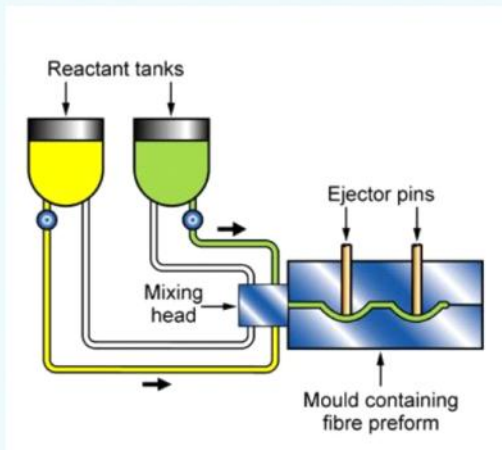
Which process is this?



Select one:

- ☐ a. Hot Pressing
- ☐ b. Hot Isostatic Pressing
- ☒ c. Sintering
- ☐ d. Tape Casting

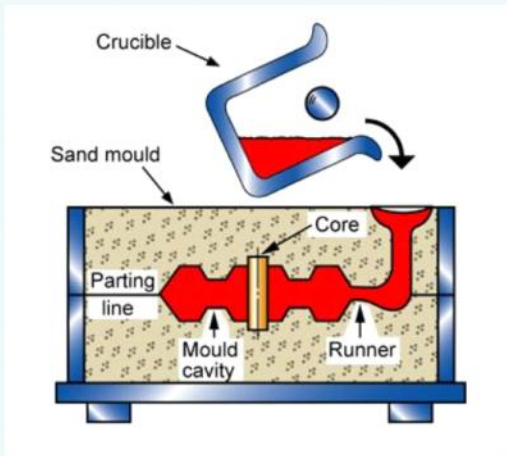
Which process is this?



Select one:

- ☐ a. Hand Lay Up
- ☐ b. Pultrusion
- ☐ c. Autoclave Moulding
- ☒ d. Resin Transfer Moulding

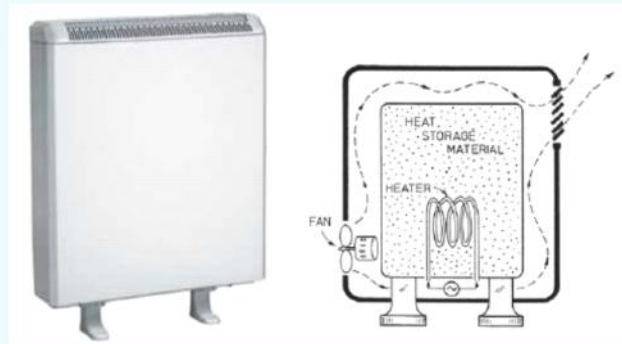
Which process is this?



Select one:

- ☐ a. Die Casting
- ☐ b. Investment Casting
- ☐ c. Uphill teaming
- ☒ d. Sand Casting

The demand for electricity is greater during the day than during the night. It is not economic for some electricity generation facilities to reduce output, so they seek instead to smooth demand by charging less for off-peak electricity. Cheap, off-peak electricity can be exploited for home or office heating by using it to heat a large mass of thermal storage material from which heat is later extracted during off-peak hours by blowing air at a controlled rate over the hot mass.



For an overnight storage heater the main aim is to maximise the heat stored per unit cost. The first thing to consider is the energy storage capacity of different materials. What is the formula for the amount of energy, Q , stored in a mass, m , of material with a specific heat capacity, C_p , when heated through a temperature interval δT is

Select one:

- ☐ $Q = m^2 C_p^2 \delta T$
- ☒ $Q = m C_p \delta T$
- ☐ $Q = \frac{m}{C_p \delta T}$
- ☐ $Q = \frac{m C_p}{\delta T}$

$$Q = m C_p \delta T \quad (\text{Energy stored in material})$$

For the same mass of material, what would the cost, C , be if the cost per unit mass of the material is C_m ?

Select one:

- ☐ $C = \log \frac{1}{m C_m}$
- ☒ $C = m C_m$
- ☐ $C = \frac{C_m}{m}$
- ☐ $C = \frac{m}{C_m}$

$$\left. \begin{array}{l} C_m: \text{Units} = \text{£/Kg} \\ m: \text{Units} = \text{Kg} \end{array} \right\} \text{Cost, } C = m \times C_m \quad (\text{Units} = \text{£})$$

If you divide the two parameters just calculated then you can calculate the amount of heat that can be stored per unit cost as:

Select one:

- ☐ $\frac{Q}{C} = \frac{C_m}{C_p} \delta T$
- ☒ $\frac{Q}{C} = \frac{C_p}{C_m} \delta T$
- ☐ $\frac{Q}{C} = C_p \cdot (C_m)^2 \delta T$
- ☐ $\frac{Q}{C} = C_p \cdot C_m \delta T$

$$\frac{Q}{C} = \frac{\cancel{m} C_p \delta T}{\cancel{m} C_m} = \frac{C_p}{C_m} \delta T$$

For a specified δT , the index that we want to maximise is therefore the maximum amount of heat that can be stored per unit cost. What is the materials index that we would like to maximise

Select one:

- ☐ $M_1 = C_p \cdot C_m$
- ☒ $M_1 = \frac{C_p}{C_m}$
- ☐ $M_1 = \frac{C_m}{C_p}$
- ☐ $M_1 = C_p \cdot (C_m)^2$

$$\frac{Q}{C} = \frac{C_p}{C_m}$$

$$\text{Maximise } \frac{Q}{C} \Rightarrow \text{Maximise } \frac{C_p}{C_m}$$

Looking through your calculations, determine what additional materials requirement that you might want to impose in order to maximise the energy storage capacity of the materials that you have chosen for this application?

Select one:

- ☐ a. You want the material selected to have a low value for its lowest value of the maximum service temperature.
- ☐ b. You want the material selected to have a low value for its maximum value of the maximum service temperature.
- ☒ c. You want the material selected to have a high value for its lowest value of the maximum service temperature.
- ☐ d. You want the material selected to have a high value for its maximum value of the minimum service temperature.